

ASSIGNMENT -2.4

Q1. Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

Ans:-

* **Steps to Connect and Test a USB Webcam**

- **Connect the Webcam:** Plug the USB webcam into an available USB port on your computer.
- **Wait for Detection:** Windows should automatically detect the webcam and install the required driver.
- **Install Drivers if Required:** If Windows does not recognize the webcam, download and install the correct driver from the webcam manufacturer's official website.
- **Open the Camera App:** Click **Start** → **Camera** to open the built-in Windows Camera application.
- **Check the Video:** Make sure your webcam image appears on the screen. If necessary, select the USB webcam from the camera settings.
- **Allow Permissions:** If prompted, allow the Camera app to access the webcam.
- **Capture a Photo:** Click the **Take Photo** button in the Camera app.
- **Verify the Photo:** Open the captured image and check that the photo is clear and the webcam is working correctly.

Q2. Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

Ans:-

* **Steps to Set Up and Test a Local Printer**

- **Connect the Printer:**

Connect the printer to the computer using a **USB cable**. If it is a network printer, connect both the printer and computer to the same **Wi-Fi or wired network**.

- **Turn on the Printer:**

Switch on the printer and make sure it has paper and sufficient ink or toner.

- **Open Printer Settings:**

On Windows, go to **Settings → Bluetooth & devices → Printers & scanners**.

- **Add the Printer:**

Click **Add device** and wait for Windows to detect the printer. Select the correct printer and follow the on-screen instructions.

- **Install the Driver:**

Windows may automatically install the required driver. If it does not, download the correct driver for the printer model from the manufacturer's official website and install it.

- **Set as Default Printer:**

After installation, select the printer and choose **Set as default** if you want to use it as your main printer.

- **Print a Test Page:**

Open the printer's settings, select **Printer properties**, and click **Print Test Page**.

- **Check the Output:**

Verify that the test page prints correctly without missing text, lines, or major errors.

Q3. Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem.

Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

Ans:-

Steps

- First, switch on the printer and connect it to the computer using a **USB cable**.
- Confirm that the printer is working and visible under **Settings** → **Bluetooth & devices** → **Printers & scanners**.
- To simulate a connectivity problem, **unplug the printer's USB cable** from the computer.
- Press **Windows + X** and select **Device Manager**.
- In Device Manager, check **Printers**, **Print queues**, and **Universal Serial Bus controllers**.
- Look for a **yellow warning triangle (!)** or an **Unknown Device**. This may indicate that Windows cannot properly detect the printer or its driver.
- Reconnect the printer's USB cable firmly to the computer.
- In Device Manager, click **Action** → **Scan for hardware changes**. Windows will search for newly connected hardware and should detect the printer again.
- If the printer still does not appear, right-click the related device and select **Update driver** → **Search automatically for drivers**.
- If necessary, try another USB port and restart the computer.
- Finally, open **Settings** → **Bluetooth & devices** → **Printers & scanners** and confirm that the printer is listed.
- Print a **test page** to verify that the problem has been successfully resolved.

Q4. A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications.

Hint: Check if another app is using the webcam or if Zoom permissions are enabled.

Ans:-

Troubleshooting Webcam Not Working in Zoom

Steps to Troubleshoot and Resolve

- **Close Other Webcam Applications**

Close the Windows Camera app and any other application that may be using the webcam. Sometimes, only one application can access the camera at a time.

- **Restart Zoom**

Completely close Zoom and open it again. Then join a meeting and check whether the webcam video appears.

- **Check Zoom Camera Selection**

In Zoom, go to **Settings** → **Video** and check the **Camera** option. Select the correct USB webcam from the list.

- **Check Camera Permissions**

Open **Windows Settings** → **Privacy & security** → **Camera**. Make sure:

- **Camera access** is turned on.
- **Let apps access your camera** is enabled.
- **Let desktop apps access your camera** is enabled, because Zoom is a desktop application.

- **Check the Webcam in Device Manager**

Press **Windows + X** → **Device Manager**. Expand **Cameras** or **Imaging devices** and check whether the webcam is listed.

- **Check for Error Icons**

Look for a yellow warning symbol next to the webcam. If present, right-click the webcam and select **Update driver**.

- **Reconnect the Webcam**

If it is a USB webcam, unplug it and connect it again. Try another USB port if necessary.

- **Restart the Computer**

Restart Windows and open Zoom again. This can clear temporary camera-access conflicts.

- **Test the Webcam in Zoom**

Open **Zoom** → **Settings** → **Video** and confirm that the webcam preview is now visible.